

Interface Configuration

At the end of this episode, I will be able to:

1. Interface Configuration

Leamer Objective: *Learn about the technologies that are typically configured when working with switch interfaces*

Description: In this episode, you will learn about configuring switch interfaces in a typical modern network. These technologies you often configure include the native VLAN, the voice VLAN, 802.1Q tagging, link aggregation, speed, and duplex.

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- The native VLAN is the default VLAN on a network that carries untagged traffic. It is commonly used for communication between devices that are not configured to participate in VLANs or for management traffic on a trunk link.
 - Voice VLAN is a network configuration feature that allows the segregation of voice and data traffic on a single physical network, typically implemented in Ethernet networks. It enables the prioritization and separate handling of voice over IP (VoIP) traffic, ensuring better quality of service for voice communications within an organization's network infrastructure.
 - 802.1Q tagging is a standardized protocol in computer networking that enables the insertion of additional information, known as VLAN tags, into Ethernet frames. This tagging allows for the identification and segregation of different virtual LANs (VLANs) within a single physical network, facilitating efficient traffic management and network segmentation.
 - Link aggregation, also known as port trunking or bonding, is a networking technique that combines multiple parallel network connections between devices to operate as a single, higher-bandwidth link. This method enhances both the speed and fault tolerance of the network connection, providing increased throughput and improved resilience by distributing traffic across multiple physical links.
 - Switch interface link speed refers to the data transfer rate supported by a network switch port, typically measured in megabits per second (Mbps) or gigabits per second (Gbps). The link speed represents the maximum capacity at which data can be transmitted between the switch and the connected device, ensuring efficient and high-speed communication within the network.
 - Switch interface duplex setting refers to the communication mode between network devices connected through a switch port, determining whether data transmission is bidirectional or unidirectional. It can be set to "half-duplex," allowing communication in one direction at a time, or "full-duplex," enabling simultaneous two-way communication, optimizing network performance by minimizing collisions and enhancing overall data throughput.
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Additional Resources:

Configuring Switch Interfaces

[https://content.cisco.com/chapter.sjs?](https://content.cisco.com/chapter.sjs?uri=/searchable/chapter/www.cisco.com/content/en/us/td/docs/switches/lan/catalyst2950/software/release/12-1_11_yj4/configuration/guide/lrescg/swint.html.xml)

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